



MICROCHIP

**Steps To Modify And Work
With The CAN BUS Analyzer
GUI/Firmware Project**



Agenda

Setting Up PC Environment For GUI And Firmware

PC GUI

- Visual C++ 2008 Express **
- .NET Framework
- WDK

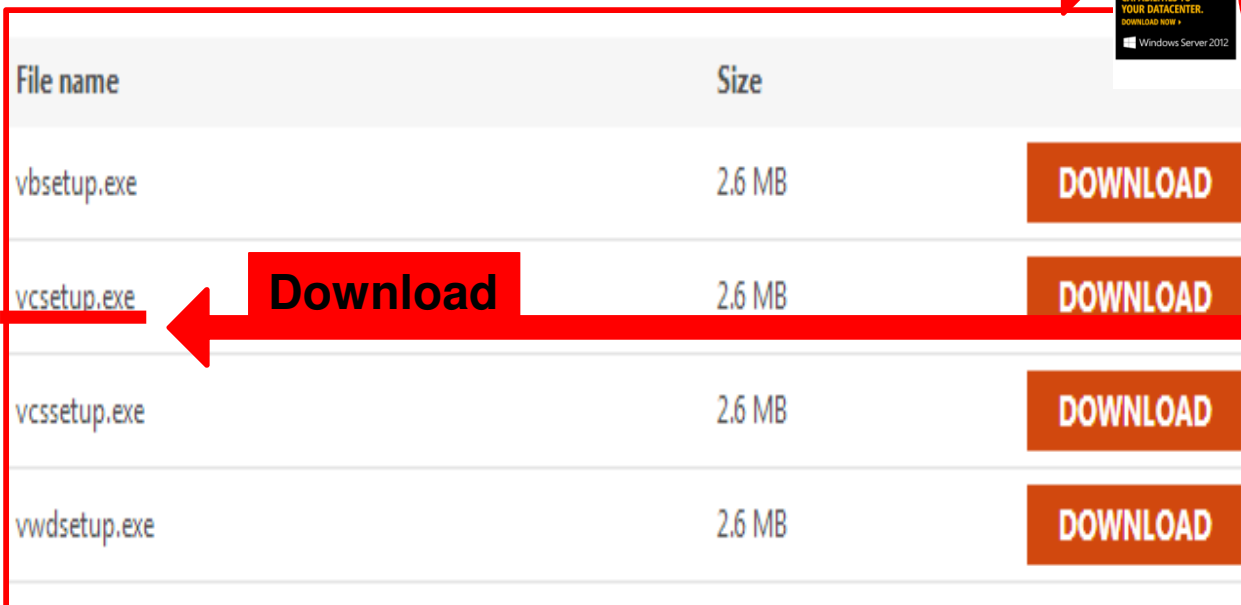
PIC18F2550 & PIC18F2680 Firmware

- MPLAB 8.66 (or higher)
- MCC3.35 (or higher)

** Downloading Visual C++ 2008 will also download an acceptable version of the .NET Framework needed to modify the source code.



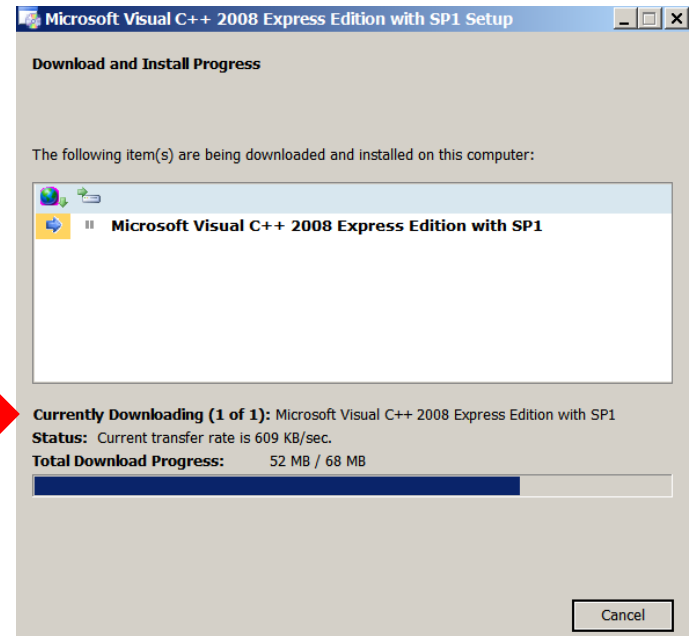
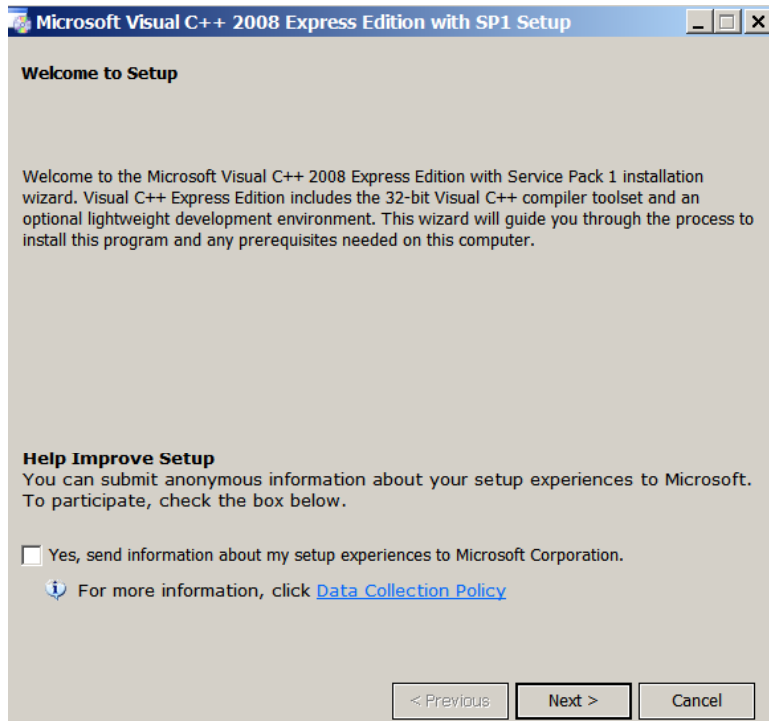
Go to: <http://www.microsoft.com/en-us/download/details.aspx?id=14597>





Installing Visual C++

Follow the setup instructions to download



Finished!



Downloading The WDK

- Download the latest WDK

Go to: <http://www.microsoft.com/en-us/download/details.aspx?id=11800>

Burn the .ISO file to a CD and run it on your computer to download the WDK

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Windows Driver Kit Version 7.1.0

Quick links

- Overview
- System requirements
- Instructions

The Windows Driver Kit (WDK) Version 7.1.0 is an update to the WDK 7.0.0 release and contains the tools, code samples, documentation, compilers, headers and libraries with which software developers create drivers for Windows 7, Windows Vista, Windows XP, Windows Server 2008 R2, Windows Server 2008, and Windows Server 2003.

Quick details

Version:	7.1.0	Date published:	2/26/2010
Language:	English		
File name		Size	
GRMWDK_EN_7600_1.ISO		619.8 MB	

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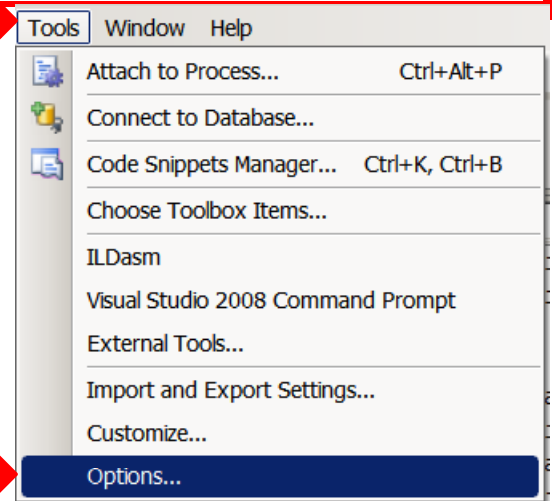
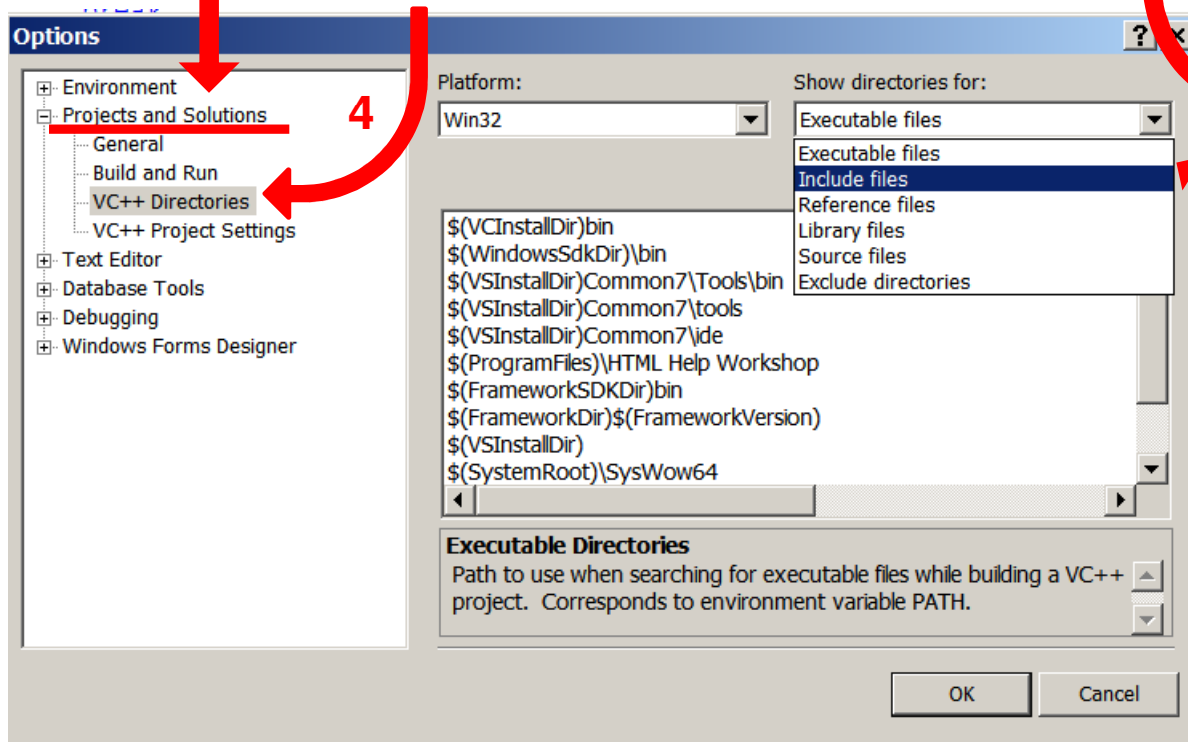
Linking The WDK To VC++

To link the WDK to the VC++ IDE:

Open VC++ and select 'Tools'
Then select 'Options...'

Go to 'Projects and Solutions'

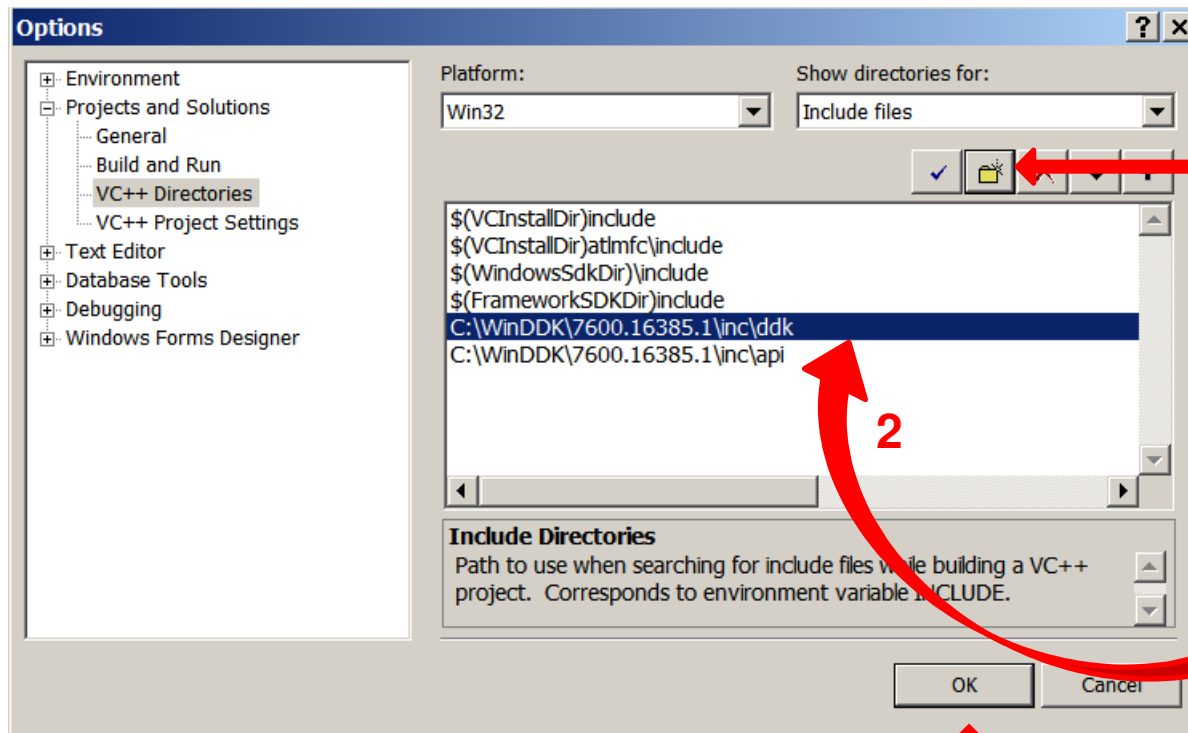
3 Choose 'VC++ Directories'



5 Select 'Include files'



Linking The WDK To VC++



Click on the new file icon

1

It will create a new line to edit.

Type in C:\WinDDK\ [add the proper version #] \inc\ddk and C:\WinDDK\ [proper version #] \inc\api

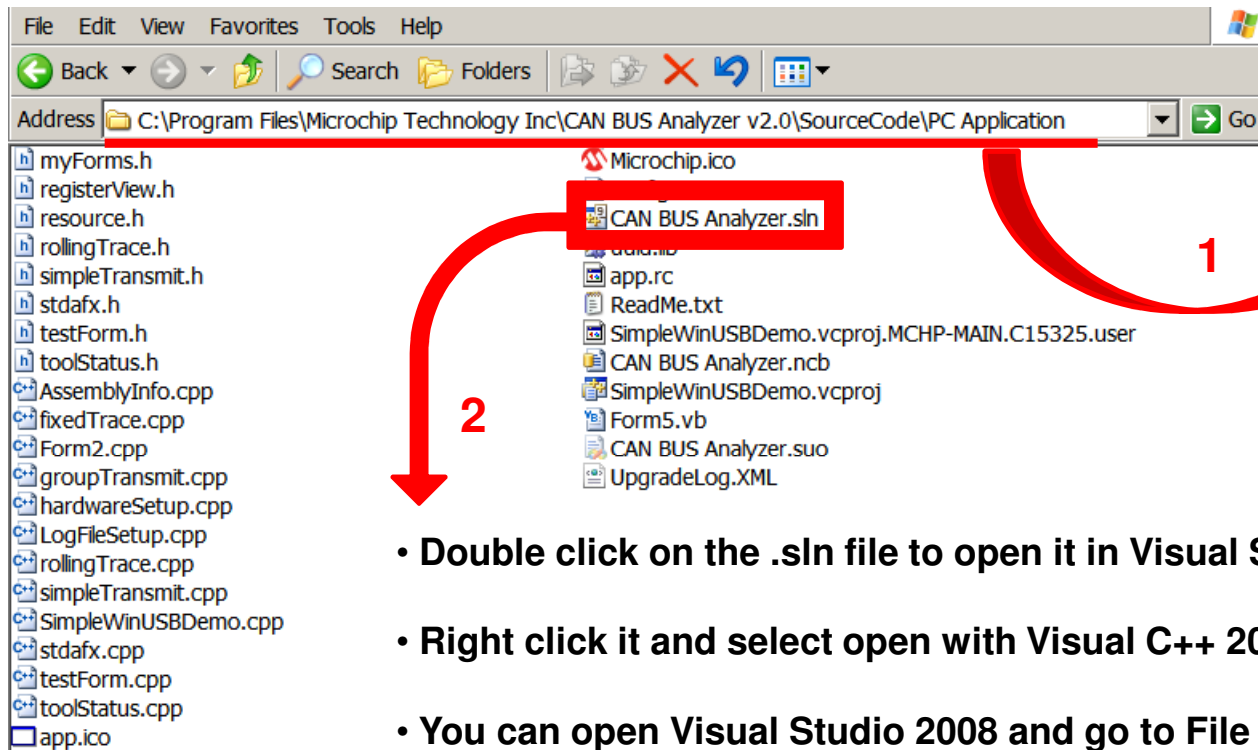
In that order at the bottom of the list.

3

Then click 'OK' and your IDE should be configured.



Opening The GUI Project



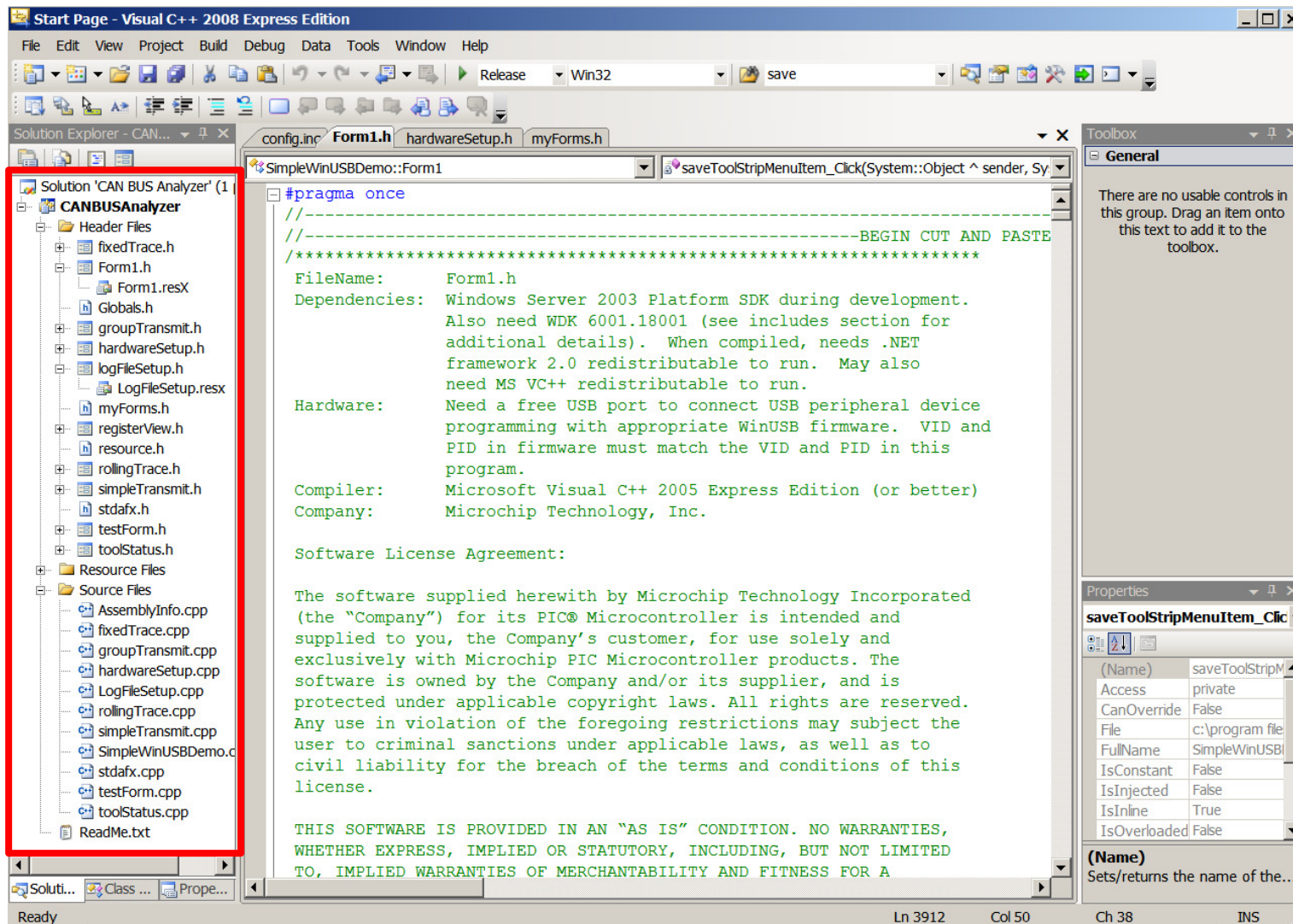
Open the folder containing the solution file.

- Double click on the .sln file to open it in Visual Studio 2008.
- Right click it and select open with Visual C++ 2008 Express.
- You can open Visual Studio 2008 and go to File > Open > Project/Solution and search for the file inside that window to select it and open it.
- Or if you have worked with the project before, open Visual Studio 2008 and select CAN BUS Analyzer in the 'Recent Projects' list in the top left corner.

C:\Program Files\Microchip Technology Inc\CAN BUS Analyzer v2.0\SourceCode\PC Application)



Modifying The GUI Project



**CAN BUS
Analyzer
opened up
in Visual
C++ 2008**



Downloading MPLAB 8.66

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Downloading MPLAB 8.66

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MPLAB IDE v5.70.40	MPLAB IDE v6.20	MPLAB IDE v7.10
	MPLAB IDE v6.30	Patch for MPLAB IDE v7.10
	MPLAB IDE v6.40	MPLAB IDE v7.20
	MPLAB IDE v6.50	MPLAB IDE v7.30
	MPLAB IDE v6.60	MPLAB IDE v7.31
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		MPLAB IDE v7.50
		MPLAB IDE v7.60
		MPLAB IDE v8.00
		MPLAB IDE v8.10
		MPLAB IDE v8.14
		MPLAB IDE v8.15
		MPLAB IDE v8.20a
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[MPLAB IDE v8.53](#)

[MPLAB IDE v8.56](#)

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[MPLAB IDE v8.63](#)

[MPLAB IDE v8.66](#)

[MPLAB IDE v8.70](#)

[MPLAB IDE v8.73a](#)

[MPLAB IDE v8.76](#)

[MPLAB IDE v8.80](#)



Installing MPLAB 8.66

1

2

3

MPLAB_IDE_v8_...

Open
Search...
Explore
Extract All...
Save Desktop Icon Layout
Restore Desktop Icon Layout
Scan for Viruses...
7-Zip
Open With
Send To
Cut
Copy
Create Shortcut
Delete
Rename
Properties

setup.exe
ISSetup.dll
mplabcert.bmp
Data1.cab
MPLAB Tools v8.66.msi

MPLAB@ Tools

Setup Status

MICROCHIP

MPLAB Tools v8.66 is configuring your new software installation.

Validating install

Cancel



Downloading MCC 3.35

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		MPLAB IDE v7.40
		MPLAB IDE v7.50
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		MPLAB IDE v8.10
		MPLAB IDE v8.14
		MPLAB IDE v8.15
		MPLAB IDE v8.20a
		MPLAB IDE v8.30
		MPLAB IDE v8.33
		MPLAB IDE v8.36

Scroll Down

Find:
MPLAB C Compiler
for PIC18

MPLAB C Compiler for PIC18

[v2.40 upgrade](#)

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[v3.16 upgrade](#)

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[v3.33 upgrade](#)

[v3.35 upgrade](#)

[v3.37.01 upgrade](#)

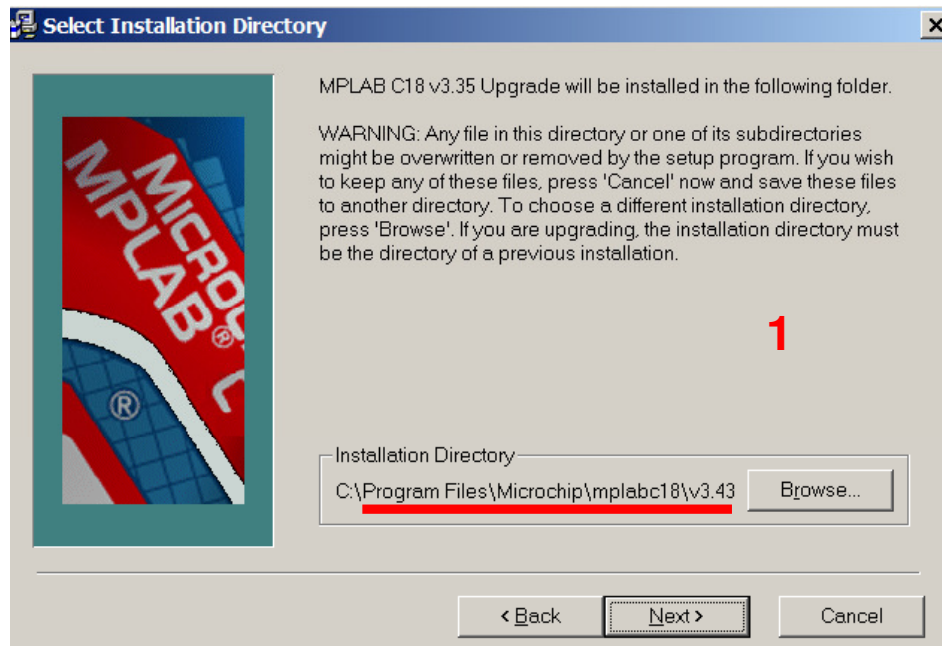
[v3.40 upgrade](#)

[v3.42 upgrade](#)

[v3.44 upgrade](#)

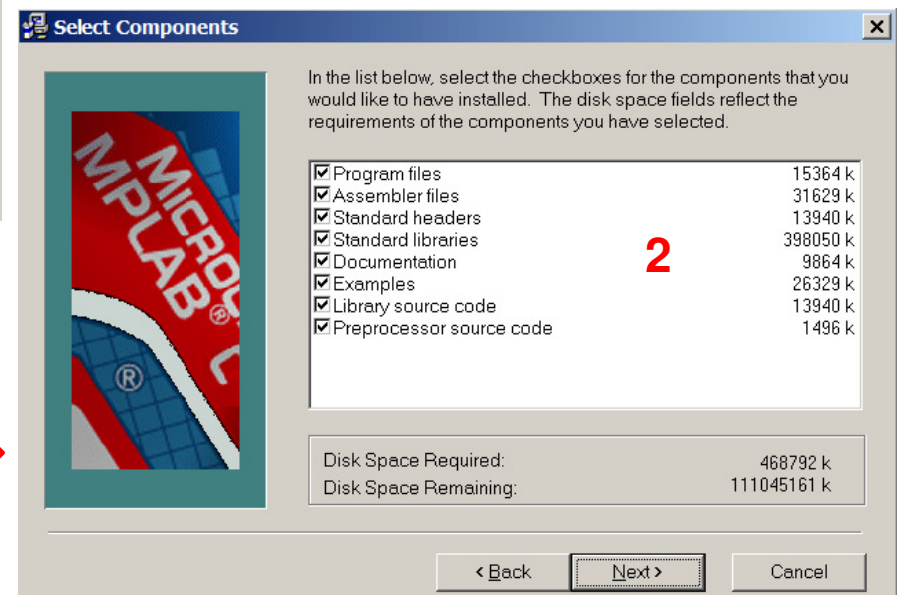


Installing MCC 3.35



Run the executable that you downloaded and follow the installation instructions shown to save to the path show.

More information is provided about the compiler, which you can pick and choose when downloading.



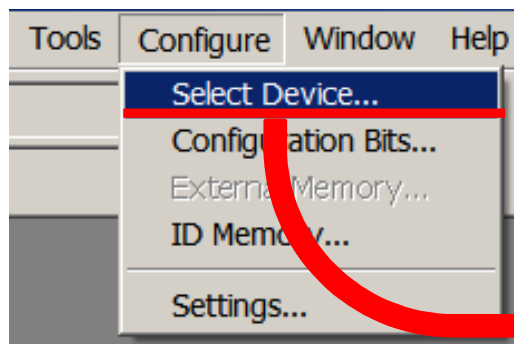


Applying the Firmware

If you need to revert back to the original firmware for whatever reasons follow these steps:

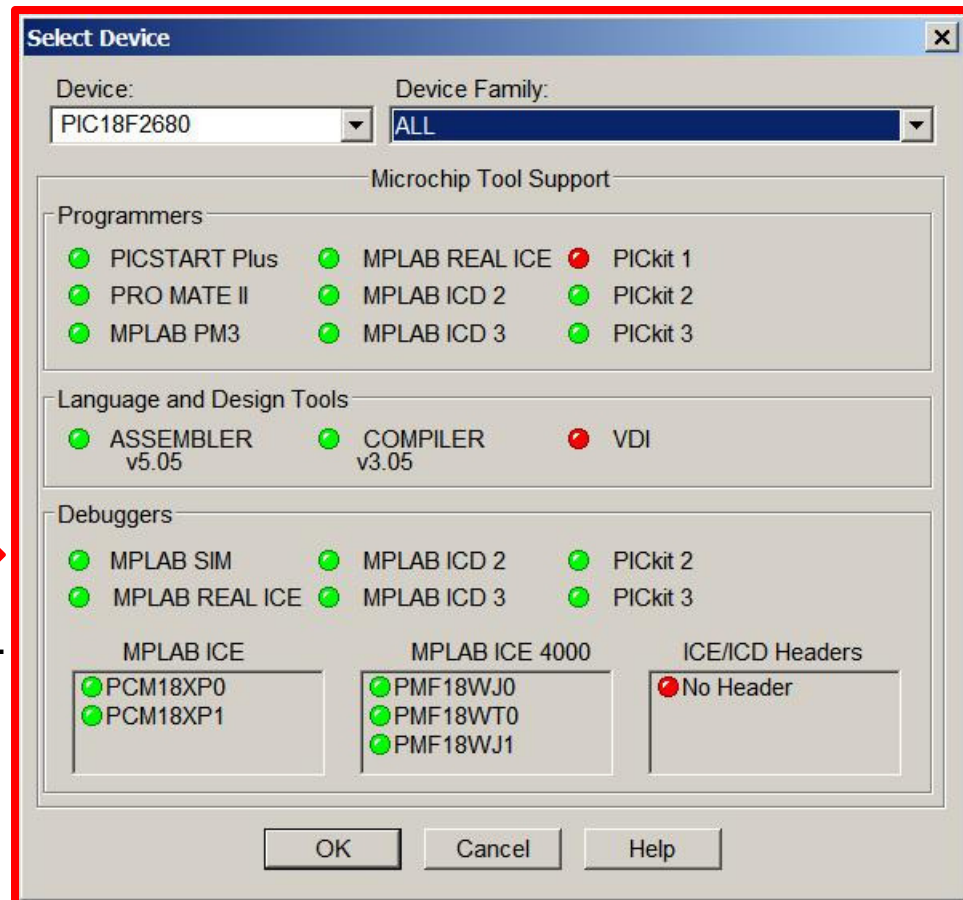
Step 1) Open MPLAB 8

Step 2) Select Device



Select the device you are programming.

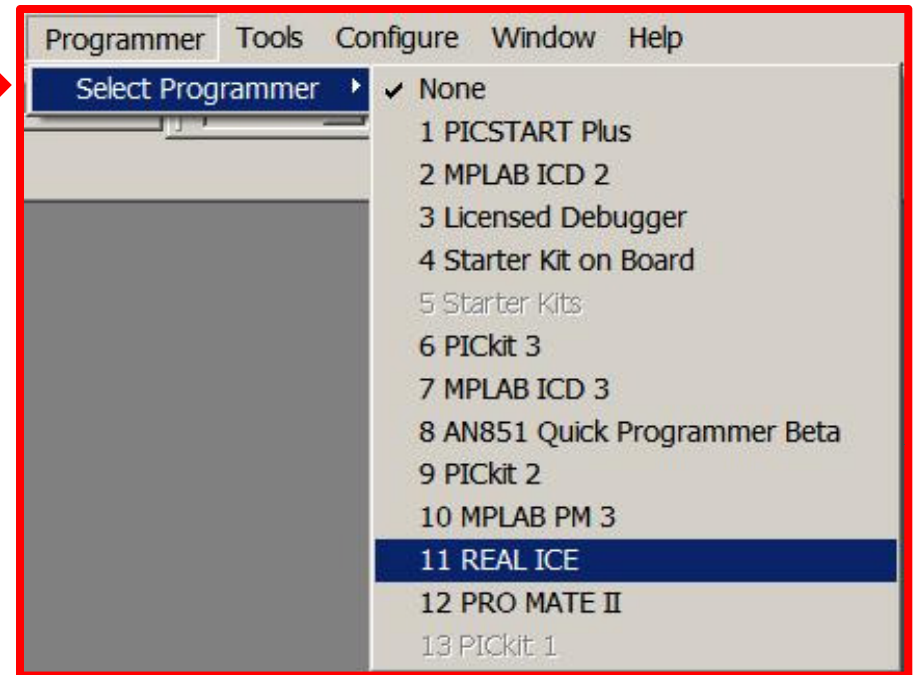
There are two micros on the CBA that can be programmed: PIC18F2680 and PIC18F2550.





Applying the Firmware

Step 3) Select the programmer



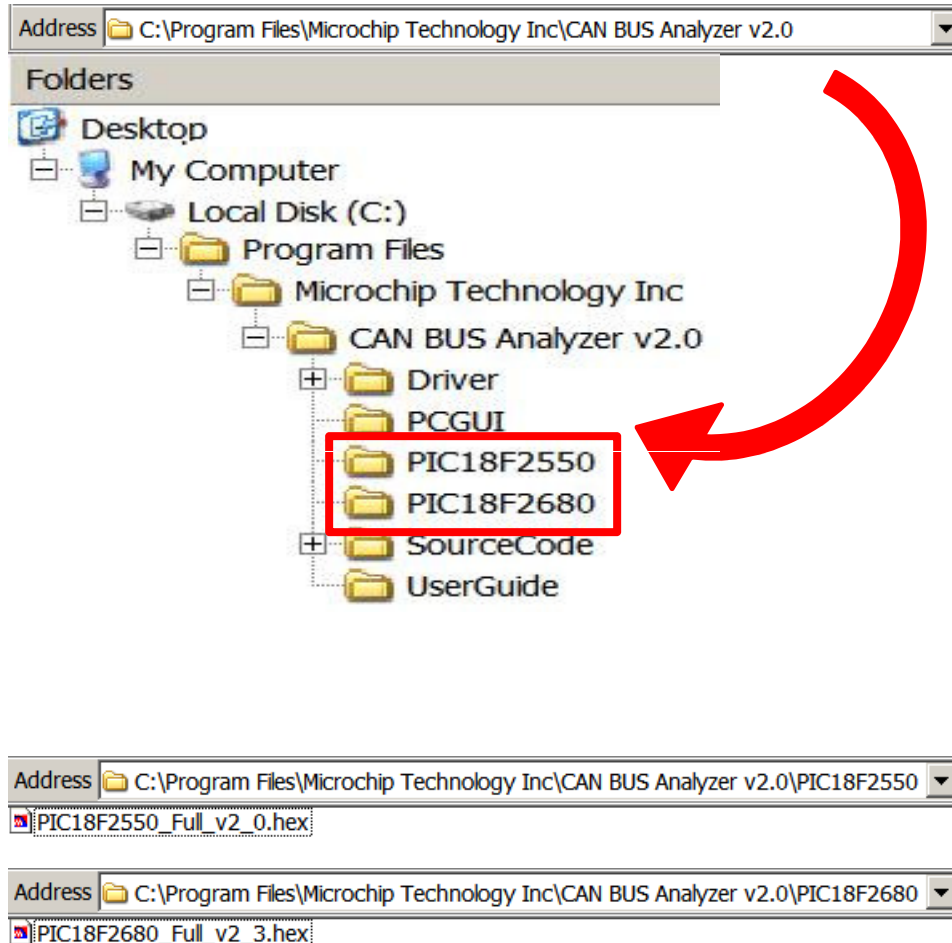
Step 4) Import the hex file



If you do not “import” the hex file you risk changing the configuration bits.



Applying the Firmware



Step 4) Import the hex file cont.

```
...
:10003000FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFD0
:10004000FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFC0
:10005000FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFB0
:10006000FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFA0
:10007000FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF90
:10008000FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF80
:10009000FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF70
:1000A000FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF60
:1000B000FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF50
:1000C000FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF40
:1000D000FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF30
:1000E000FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF20
:1000F000FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF10
:10010000FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
...
```

Can find a list of programming tools at <http://microchip.com/>



Resources

<http://msdn.microsoft.com/en-us/vstudio/hh386302.aspx>

The screenshot shows the Visual Studio website. A red arrow points from the URL above to the 'Visual C++ Resources' link in the navigation bar. The page content includes sections for 'What's New with Visual C++', 'Learn Visual C++', and 'Get Visual Studio'.

Visual Studio

Search Visual Studio with Bing

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Visual C++ Resources

What's New with Visual C++

- Julia Fractal Using C++ AMP
Check out this C++ AMP implementation of a fractal generator that renders 4-dimensional Quaternion J... [more](#)
- C++/CX Part 2 of [n]: Types That Wear Hats
Read about hat types, which manage lifetime of Windows Runtime objects and provide type conversion.

Learn Visual C++

- Get Started with Visual C++
Find the essential resources for learning Visual C++, including features, supported platforms, and an overview of Windows programming.
- Explore the Visual Studio IDE for C++
Learn how to create a standard C++ program and test its functionality by using features in Visual Studio.
- How Do I: Create a Console Application with Visual C++ Express Edition? (2005)
Create console applications using the C and C++ languages.
- Develop a Win32 Application in Visual C++ (2005)
Learn how to build a Win32 application with a graphical user interface.
- Principles of Object-Oriented Programming
Learn the principles of object-oriented programming, and write a small MFC program to

Get Visual C++

- Get Visual C++ in Visual Studio Express 2012 for Windows Desktop

Other Visual C++ Resources

- Visual C++ "How Do I?" Videos
- Visual C++ Books
- Hilo: Windows 7 C++ Development
- Developing Windows Applications in C++

Get Visual Studio

- Visual Studio 2012 Trial Download
- MSDN Subscriber Downloads
- Visual Studio 2010 Service Pack 1
- Visual Studio 2010 SDK

Development Platforms

- Windows
- Windows Phone
- Windows Embedded



Resources

Visual C++ Programming Help:



→ <http://stackoverflow.com/>



→ <http://www.cplusplus.com/>



→ <http://cboard.cprogramming.com/forum.php>

Otherwise:



→ <https://www.google.com/>